Circuits and Systems-II
ECE-1201

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“My lord, increase me in knowledge”.

[Sura Ta Ha]

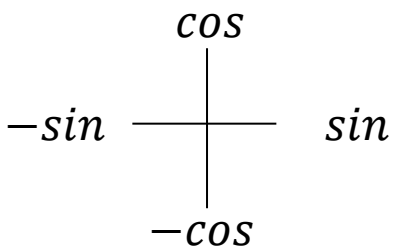
Determining Phase Sequence

Find the phase sequence for the following 3 phase system

$$v_{an} = -200\cos(\omega t)$$
$$= 200\sin(\omega t + 90°)$$

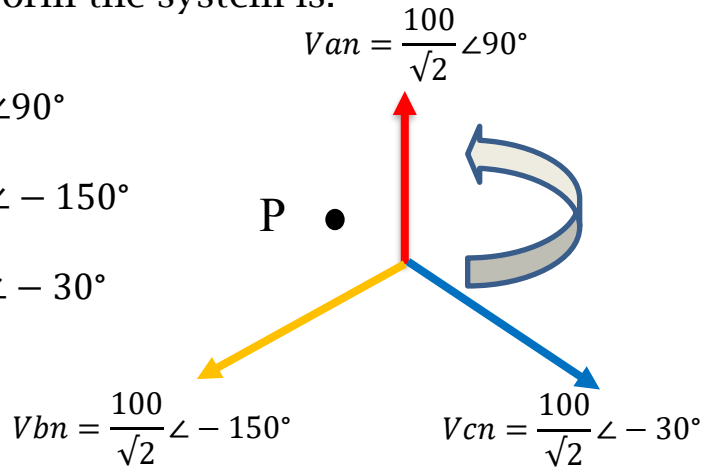
$$v_{bn} = 200\sin(\omega t - 150°)$$

$$v_{cn} = 200\sin(\omega t - 30°)$$



In phasor form the system is:

$$Van = \frac{200}{\sqrt{2}}\angle 90°$$
$$Vbn = \frac{200}{\sqrt{2}}\angle -150°$$
$$Vcn = \frac{200}{\sqrt{2}}\angle -30°$$



So phase sequence is a,c,b

Phasor summation in any situation of a balanced three phase voltage / current is always zero

Proof 1:

Assuming **positive** phase sequence [when $\omega t = 0°$]:

$$v_{an} = V_m \sin(\omega t) = 0$$

$$v_{bn} = V_m \sin(\omega t - 120°) = \frac{-\sqrt{3}}{2} V_m$$

$$v_{cn} = V_m \sin(\omega t + 120°) = \frac{\sqrt{3}}{2} V_m$$

$$i_{an} = \frac{I_m}{Z} \sin(\omega t) = 0$$

$$i_{bn} = \frac{I_m}{Z} \sin(\omega t - 120°) = \frac{-\sqrt{3}}{2} \frac{I_m}{Z}$$

$$i_{cn} = \frac{I_m}{Z} \sin(\omega t + 120°) = \frac{\sqrt{3}}{2} \frac{I_m}{Z}$$

$$v_T = v_{an} + v_{bn} + v_{cn} = 0 - \frac{\sqrt{3}}{2} V_m + \frac{\sqrt{3}}{2} V_m = 0$$

$$i_T = i_{an} + i_{bn} + i_{cn} = 0 - \frac{\sqrt{3}}{2} \frac{I_m}{Z} + \frac{\sqrt{3}}{2} \frac{I_m}{Z} = 0$$

Phasor summation in any situation of a balanced three phase voltage / current is always zero

Proof 2:

Assuming **negative** phase sequence [when $\omega t = 90°$]:

$$v_{an} = V_m \sin(90°) = V_m$$
$$v_{bn} = V_m \sin(90° + 120°) = \frac{-1}{2} V_m$$
$$v_{cn} = V_m \sin(90° - 120°) = \frac{-1}{2} V_m$$

$$v_T = v_{an} + v_{bn} + v_{cn} = V_m - \frac{1}{2} V_m - \frac{1}{2} V_m = 0$$

$$i_{an} = \frac{I_m}{Z} \sin(90°) = \frac{I_m}{Z}$$
$$i_{bn} = \frac{I_m}{Z} \sin(90° + 120°) = -\frac{1}{2} \frac{I_m}{Z}$$
$$i_{cn} = \frac{I_m}{Z} \sin(90° - 120°) = -\frac{1}{2} \frac{I_m}{Z}$$

$$i_T = i_{an} + i_{bn} + i_{cn} = \frac{I_m}{Z} - \frac{1}{2} \frac{I_m}{Z} - \frac{1}{2} \frac{I_m}{Z} = 0$$

Similarly, it can be proved for any angle for both the positive and negative sequence.

[practice randomly]

Three phase systems

As both the three-phase source and the three-phase load can be either wye- or delta-connected, there are four possible connections:

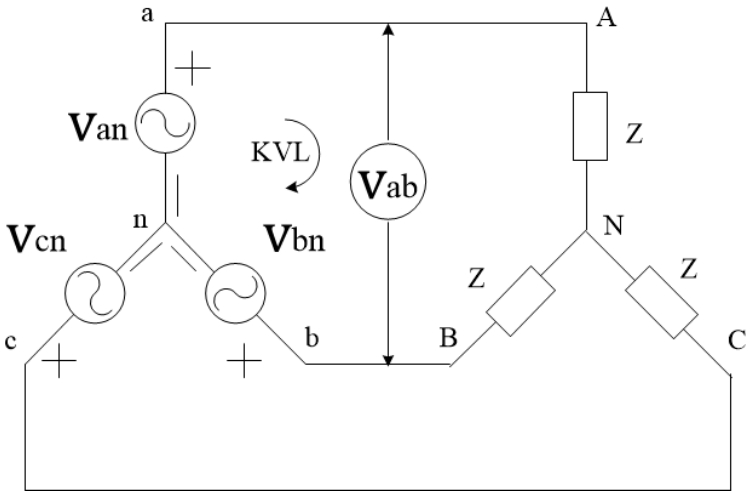
- Y-Y connection (i.e., Y-connected source with a Y- connected load)
- Y- Δ connection (i.e., Y-connected source with a Δ - connected load)
- Δ- Δ connection (i.e., Δ -connected source with a Δ - connected load)
- Δ-Y connection (i.e., Δ -connected source with a Y - connected load)

It is appropriate to mention here that a balanced **delta-connected load** is more common than a balanced wye-connected load. This is due to the ease with which loads may be added or removed from each phase of a delta-connected load. This is very difficult with a wye-connected load because the **neutral may not be accessible**. On the other hand, delta connected sources are not common in practice because of the circulating current that will result in the delta-mesh if the three-phase voltages are slightly unbalanced.

Balanced Y-Y system

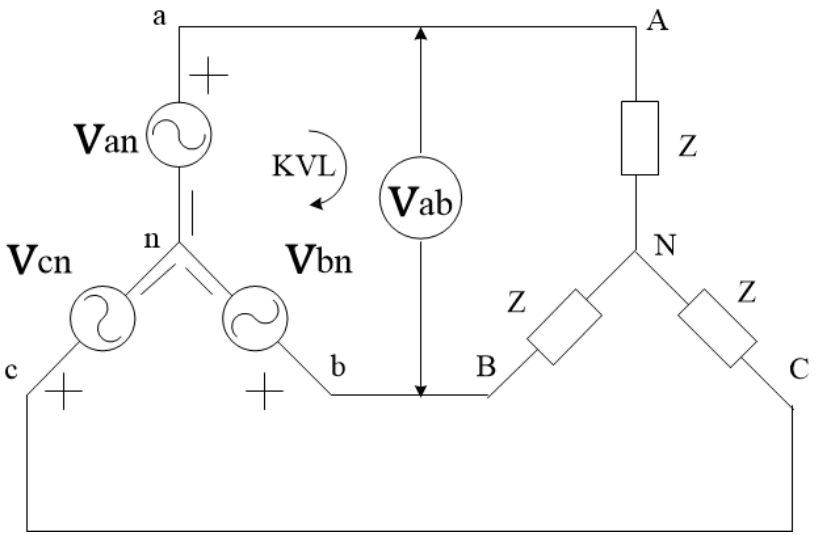
Relation between line voltage and phase voltage:

- The three **conductors** connected from a, b and c to the load are called **lines.**
- The voltage from one line to another is called a **line voltage.** [V_{ab}]
- The voltage across any phase is called a **phase voltage.** [V_{an}]
- For the Y-connected system, **line current** equals the **phase current** for each phase.



Balanced Y-Y system

Relation between line voltage and phase voltage:



Assumptions:

- *abc*/ positive/ RYB phase sequence
- Magnitude of Phase voltage =|Van|=|Vbn|=|Vcn|=V_P
- Magnitude of line voltage =|Vab|=|Vbc|=|Vca|=V_L

Applying KVL at loop 1,

$$-V_{an} + V_{ab} + V_{bn} = 0$$

$$V_{ab} = V_{an} - V_{bn}$$

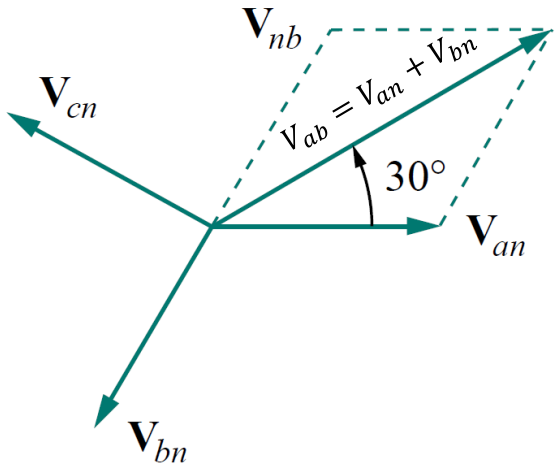
$$= V_{an} + V_{nb}$$

Balanced Y-Y system

Relation between line voltage and phase voltage:

$$V_{ab} = V_{an} - V_{bn}$$

$$= V_{an} + V_{nb} \text{ [these are vectors]}$$



$$Vab = \sqrt{V_{an}{}^2 + V_{nb}{}^2 + 2.V_{an}.V_{nb}.\cos 60°} \quad \angle 30°$$

$$V_L = \sqrt{V_P{}^2 + V_P{}^2 + 2V_P{}^2 cos60°} \quad \angle 30°$$

$$= \sqrt{V_P{}^2 + V_P{}^2 + 2V_P{}^2.\frac{1}{2}} \quad \angle 30°$$

$$= \sqrt{3}V_P \angle 30°$$

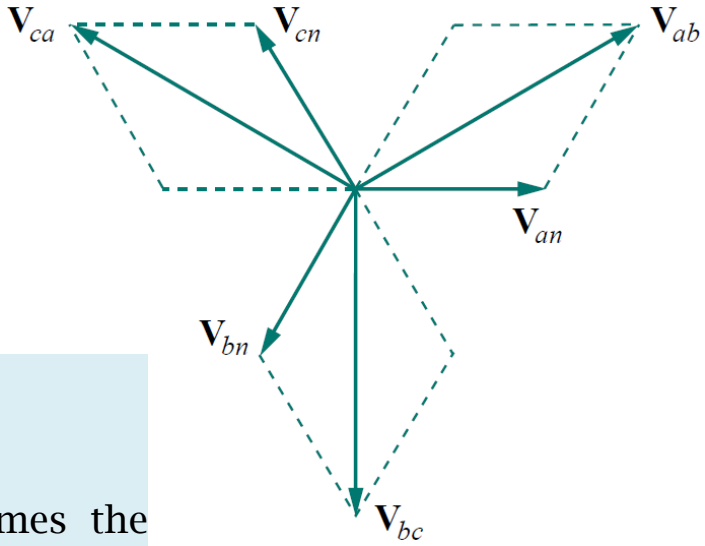
Balanced Y-Y system

Relation between line voltage and phase voltage:

Similarly, proof that,

$V_{bc} = V_L = \sqrt{3}V_P \angle -90°$ and

$V_{ca} = V_L = \sqrt{3}V_P \angle -210°$



Important Conclusion

[For ***abc*** phase sequence]

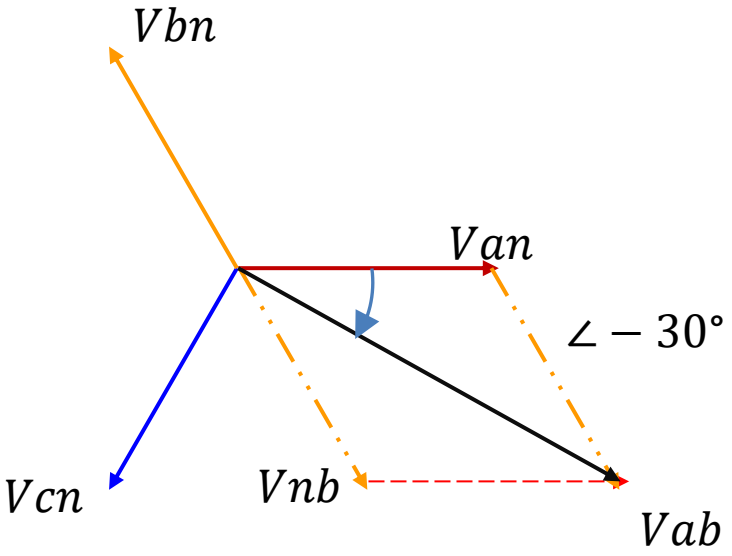
- The magnitude of the line voltages V_L is $\sqrt{3}$ times the magnitude of the phase voltages V_P
- The line voltages **lead** their corresponding phase voltages by 30°]
- **line current** equals the **phase current** for each phase.

Balanced Y-Y system

Relation between line voltage and phase voltage:

Assumptions:

- *acb*/ negative/ RBY phase sequence
- Magnitude of Phase voltage =|Van|=|Vbn|=|Vcn|=V_P
- Magnitude of line voltage =|Vab|=|Vbc|=|Vca|=V_L



$$V_{ab} = V_{an} + V_{nb}$$

$$Vab = \sqrt{{V_{an}}^2 + {V_{nb}}^2 + 2.V_{an}.V_{nb}.\cos 60°}\angle -30°$$

$$V_L = \sqrt{{V_P}^2 + {V_P}^2 + 2{V_P}^2 cos60°} \quad \angle -30°$$

$$= \sqrt{{V_P}^2 + {V_P}^2 + 2{V_P}^2.\frac{1}{2}} \quad \angle -30°$$

$$= \sqrt{3}V_P \angle -30°$$

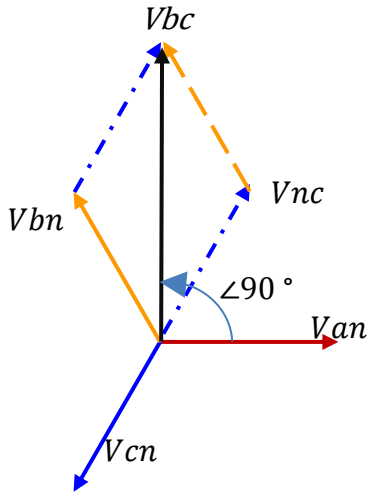
Balanced Y-Y system

Relation between line voltage and phase voltage:

Similarly, proof that,

$$V_{bc} = V_L = \sqrt{3}V_P \angle 90°$$

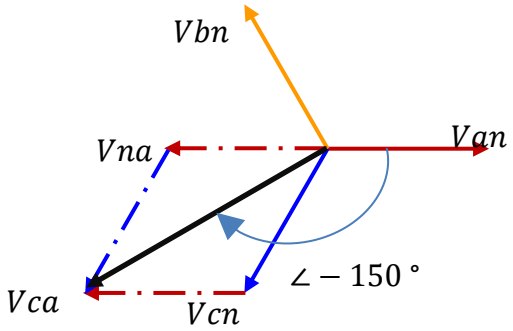
$$\text{and } V_{ca} = V_L = \sqrt{3}V_P \angle -150°$$



Important Conclusion

[For ***acb*** phase sequence]

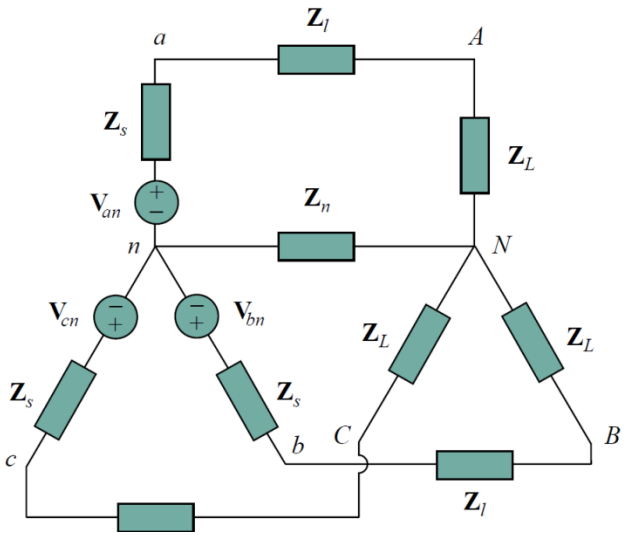
- The magnitude the magnitude of the line voltages V_L is $\sqrt{3}$ times the magnitude of the phase voltages V_P
- The line voltages **lag** their corresponding phase voltages by 30°]
- **line current** equals the **phase current** for each phase.



Contribution of neutral wire in Y-Y system

Assumptions:

- Z_s = *Source impedance*
- Z_l = *Line impedance*
- Z_L = *Load impedance*
- Z_n = *neutral wire impedance*



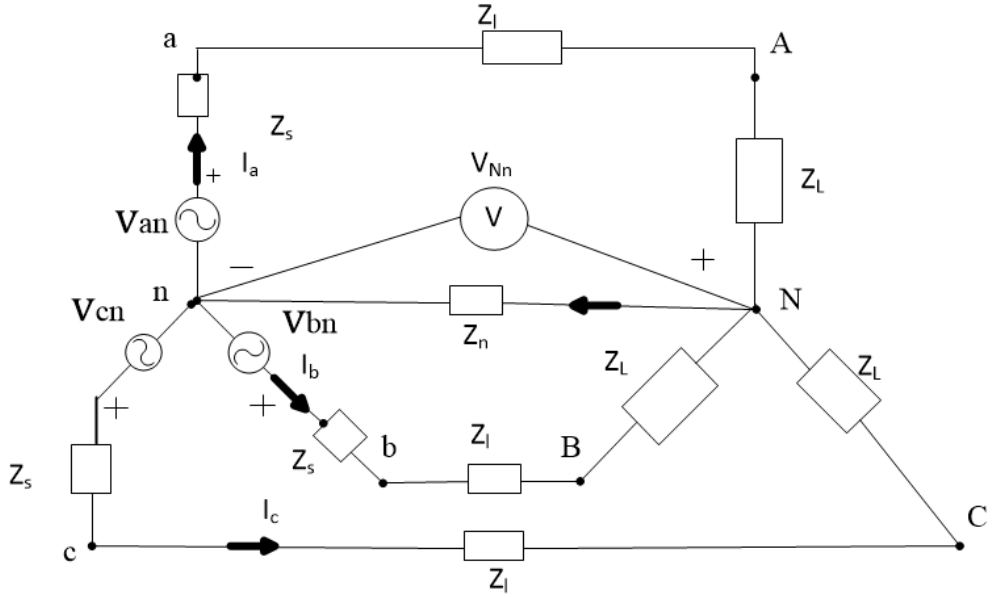
balanced load is considered so that load impedances are equal. The impedance Z_Y is the total load impedance per phase, i.e., the sum of the source impedance Z_s, line impedance Z_l, and load impedance Z_L for each phase.

$$Z_Y = Z_s + Z_l + Z_L$$

Z_s and Z_l are often very small compared with Z_L, so it can be assumed that $Z_Y = Z_L$

Contribution of neutral wire in Y-Y system

Previous figure can be redrawn as



From the loop

$-V_{an} + I_a(Z_s + Z_l + Z_L) + V_{Nn} = 0$

$V_{an} = I_a Z_Y + V_{Nn}$ (i) [For phase a]

Similarly,

For phase b and c

$V_{bn} = I_b Z_Y + V_{Nn}$ (ii) [For phase b]

$V_{cn} = I_c Z_Y + V_{Nn}$ (iii) [For phase c]

After adding equations (i), (ii) and (iii)

$$V_{an} + V_{bn} + V_{cn} = (I_a + I_b + I_c)Z_Y + 3V_{Nn}$$

$$\text{Or, } 0 = 0 \times Z_Y + 3V_{Nn}$$

$$\therefore V_{Nn} = 0$$

Also,

$$I_n = I_a + I_b + I_c = 0$$

Contribution of neutral wire in Y-Y system

In practice, if a factory, for example, had only balanced, three-phase loads, the **absence** of the **neutral** would have no effect since, ideally, the system would always be balanced. The cost would therefore be less since the number of required conductors would be reduced. However, lighting and most other electrical equipment will use only one of the phase voltages, and even if the loading is designed to be balanced (as it should be), there will never be perfect continuous balancing since lights and other electrical equipment will be turned on and off, upsetting the balanced condition. The **neutral** is therefore **necessary** to carry the **resulting current** away from the load and back to the Y-connected generator.

Practice Example and Problem 12.2 of Alexander, 22.1 of Boylstead and of similar types